

CampusSync ERP - Attendance Module Summary Report

Generated on: September 16, 2026 at 10:33 PM • System Module Documentation

Executive Summary

This report documents all technical modifications, database architecture updates, service layer additions, and UI enhancements implemented for the **CampusSync Attendance Management System**. The module encompasses both the **Faculty Manual Attendance Register** (with Draft/Final locking and dynamic semester cycle filtering) and the **Student Attendance Dashboard** (featuring circular ring gauge, subject breakdown, daily logs, and QR code scanning).

1. Faculty Attendance Register System

- **Dynamic Semester Cycle Filtering:** Binds assigned subject dropdowns directly to the College Academic Setting (AcademicSetting.semester_cycle). Automatically displays ONLY Odd Semesters (Sem 1, 3, 5) or EVEN Semesters (Sem 2, 4, 6) dynamically.
- **Save as Draft Mode:** Saves lecture attendance into database with status 'Draft'. Faculty can reload the register anytime to modify individual student P/A entries without locking the session.
- **Final Submit & Locking Safety:** Finalizes attendance with status 'Submitted'. Locks all UI controls (radio buttons, save buttons) and backend API from further edits. Automatically updates cumulative student total/attended lectures in attendance_records.
- **Interactive Controls:** Features 'Mark All Present' toggle switch, live status counter badges (Total, Present, Absent), and stylized Green 'P' / Red 'A' radio buttons.
- **QR Code Attendance Button:** Dedicated action button on the control bar for live student scanner verification.

2. Student Attendance Dashboard & QR Scanner

- **Overall Circular Donut Gauge:** SVG-rendered circular progress gauge showing overall attendance percentage color-coded by University thresholds (■ Safe ≥ 75%, ■ Warning 60-74%, ■ Shortage < 60%).
- **Subject-Wise Breakdown Cards:** Card grid displaying enrolled subjects, conducted/attended lectures, progress bar, percentage, and converted **HNGU Assessment Score (/5.0 Marks)**.
- **Daily Attendance History Table:** Displays recent 15 lecture logs showing Date, Session/Lecture No, Subject, Faculty Name, and Status badge (Present P / Absent A).
- **QR Code Attendance Scanner (/student/scan):** Integrated camera scanner route allowing students to scan active faculty QR codes and record instant Present attendance in real-time.

3. Database Architecture & Schema Summary

Table Name	Purpose & Description	Key Columns	Status / Behavior
lecture_attendance_sessions	Tracks daily lecture session headers conducted by faculty.	id, faculty_id, subject_id, semester, division, lecture_date, lecture_no, status	ENUM('Draft', 'Submitted'). Controls editing vs locking.
lecture_attendance_students	Stores individual student P/A status per lecture session.	id, session_id, student_id, status	ENUM('Present', 'Absent'). Detailed log.

attendance_records	Cumulative semester summary for total and attended lectures.	id, student_id, subject_id, total_lectures, attended_lectures	Recalculated upon Final Submit & QR scan. Used for HNGU /5 score.
--------------------	--	---	---

4. Technical Verification & Integrity Summary

- **MySQL Tables Initialized:** Created via db.create_all() and verified active in MySQL database.
- **Service Layer Integration:** Services in services/attendance_service.py and services/faculty_service.py provide atomic transactions.
- **Route Safeguards:** All endpoints enforce login and role-based decorator authorization.